EcoEyeSpy

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Abstract

In this work, we present a computer vision approach for screening, detecting, and segmenting waste deposits using an AerialWaste dataset. Our approach integrates three complementary components: (i) a ResNet50+FPN image-level classifier to pre-filter candidate scenes under severe class imbalance; (ii) a YOLOv8-based object detector tailored to small, cluttered structures typical of aerial waste imagery; and (iii) segmentation models, including both a DeepLabV3 and the Segment Anything Model (SAM) approach to delineate waste regions at the pixel level. Despite sparse supervision in the dataset, we address annotation scarcity through patch-based tiling, class-balancing strategies, and few-shot adaptation of SAM. Experiments demonstrate strong classification performance (ROC–AUC 0.88), competitive detection accuracy (mAP50 0.35), and robust segmentation, with SAM achieving a mean IoU of 0.74 under 100-shot fine-tuning. Together, these results show that modern deep learning methods, can substantially enhance automated monitoring of illegal landfills.

1. Introduction

Illegal waste disposal poses serious environmental and public-health risks, yet mapping such sites at scale remains challenging. Traditional monitoring relies on on-site inspections and citizen reports, which are costly, sparse, and often delayed, while high-resolution aerial and satellite imagery is abundant but difficult to analyse automatically because waste deposits are small, rare, and easily confused with industrial or construction structures. In this work we present EcoEyeSpy, an end-to-end computer vision pipeline for identifying and characterising waste sites in overhead imagery using the AerialWaste dataset. Our approach combines an image-level ResNet50+FPN classifier to pre-screen candidate scenes, a YOLOv8-based detector to localise individual waste instances, and segmentation models (DeepLabV3 with a ResNet-50 backbone and the Segment Anything Model) to delineate their spatial extent.

[0] https://github.com/OmarN-Mukhtar/cv_ecoeyespye

2. Related Works

Torres and Fraternali [3] introduce AerialWaste, a dataset specifically designed for discovering illegal landfills from aerial and satellite imagery. The dataset contains roughly 10k RGB tiles collected from AGEA orthophotos, WorldView-3 satellite imagery and Google Earth, curated and annotated by expert photo-interpreters. Each image is labelled at multiple levels: a binary scene label (waste vs. non-waste), richer metadata describing waste type, area type and severity of the suspected illegal activity, and pixel-wise segmentation masks for a subset of positive examples. In their work, landfill discovery is formulated as a scene-level binary classification task on full tiles, and a ResNet-50 model with a Feature Pyramid Network (FPN) backbone trained on AerialWaste achieves high average precision (around 80-85%), demonstrating that deep convolutional networks can effectively learn expert photo-interpretation patterns from overhead imagery. However, this baseline primarily exploits image-level labels and uses segmentation masks only for validation, leaving open the question of how to best leverage AerialWaste for joint detection and segmentation. In this project we build on top of that line of work by using AerialWaste not only for scene classification but also as a benchmark for localising and delineating individual waste sites, combining image-level screening with object detection and segmentation based on foundation models such as Segment Anything [2].

3. Data

3.1. AerialWaste dataset

For both classification and localisation we rely on the AerialWaste dataset from Politecnico di Milano.[1] It contains more than 10,000 high–resolution aerial and satellite images sampled over different geographic areas, some depicting illegal landfills or waste stockpiles and others showing only background. The original annotations provide (i) image–level binary labels indicating the presence or absence of waste and (ii) a more detailed taxonomy for

[1] http://aerialwaste.org

a subset of images, where waste instances are labelled by type and storage mode.

The released data are organised in COCO-style JSON files together with the corresponding image folders. Each JSON file contains an `images` table with metadata for every image (file name, spatial resolution and ancillary flags), an `annotations` table with bounding-box or segmentation labels linked by `image_id`, and a `categories` table defining the semantic classes and their higher-level `supercategory`. Overall, the dataset comprises 11,703 unique images. Only 166 images contain at least one object annotation, so 11,537 images are currently unlabeled, corresponding to a label coverage of roughly 1.4% at full-image level. This places our problem in a sparse-supervision regime: training a box-firstdetector directly on full images is not viable, and using the tiny labeled subset alone as train/validation would likely cause severe overfitting and unstable evaluation unless strong regularisation and carefully engineered splits are applied.

3.2. Label space and sampling bias

The semantic label space is organised into two main supercategories, `Type_of_object` (e.g. rubble, bulky items, tires, vehicles) and `Storage_mode` (e.g. heaps not delimited, full containers, big bags), but only a few of the more than twenty fine-grained classes are populated, with a small group of types and storage modes accounting for almost all of the 833 instances. Because of this severe imbalance we treat "waste vs background" as the primary target for detection and segmentation, collapsing rare subclasses and using the richer taxonomy mainly for descriptive analysis rather than full multi-class learning.

Image-level attributes show a similar pattern: most samples have site type, severity and evidence marked as *N/A*, and chi-square tests across these variables indicate that the annotated images are statistically different from the unlabeled bulk of the corpus. In practice, the labeled subset over-represents a limited set of site types and conditions, so models trained only on it are exposed to a restricted slice of real-world variability and their evaluation must be interpreted with this sampling bias in mind.

3.3. Segmentation masks and patch-based subset

Given that only 169 images provide pixel-wise segmentation masks, a key design question is whether such a small number is sufficient for training. While modest in absolute terms, this amount can still be workable if exploited carefully. Following suggestions from our collaborators and consistently with the dataset design in the original AerialWaste study [3], we move from full scenes to a patch-based representation: we extract fixed-size tiles from the masked images and build a new dataset of roughly 6,000 patches, balanced between patches that contain garbage and purely background patches. This substantially increases the number of supervised examples and allows the segmentation model to focus on local patterns rather than entire scenes.

4. Methods

4.1. Binary Classification

4.1.1. Data preparation and balancing

As a first modelling step, we develop an image-level binary classifier to determine whether a scene contains visible waste. For this task we rely on the COCO-style `images` table rather than the sparse bounding-box annotations described in Section 3. Each image is associated with a `severity` attribute, which we convert into a binary target: images with `is_candidate_location = 1` are labelled as waste present, while instances with a zero value are treated as background. This definition provides a consistent signal across the full dataset, including images without object-level annotations.

Because the image files are distributed across multiple directories, we implement a unified loading layer that resolves each file name, filters out missing entries, and assembles a clean metadata table. This dataset is then split into training and validation partitions using a stratified procedure to preserve the empirical class ratio.

The dataset is highly imbalanced, with positive samples representing only a small fraction of the negatives. To counter this, we employ two complementary mechanisms during training. First, we compute a positive-class weight equal to the ratio of negative to positive examples and pass it to the binary cross-entropy loss, so that misclassified positives incur a larger penalty. Second, the training dataloader uses a weighted sampling scheme in which each example receives a weight inversely proportional to its class frequency, producing batches with a more balanced composition of waste and background images.

All images are resized to 256×256 pixels and normalised using ImageNet statistics. Furthermore, using Albumentations, we apply moderate geometric and photometric augmentations, including horizontal flips, small rotations and scale changes, and brightness/contrast adjustments, to mimic realistic variation in aerial acquisition conditions.

4.1.2. Model architecture and training

The classifier is built on a ResNet-50 backbone augmented with a Feature Pyramid Network (FPN). The backbone produces feature maps at four depths (c2–c5), corresponding to its internal convolutional blocks. Using a feature extraction

interface, these intermediate activations are passed into a top–down FPN with lateral connections, which transforms all pyramid levels into 256-channel feature maps and fuses high-resolution spatial cues with deeper semantic information. This multi-scale representation is well suited to aerial imagery, where waste deposits range from small, compact piles to large diffuse structures.

Each FPN output level is fed through global average pooling, yielding four 256-dimensional descriptors. These are concatenated into a single 1024-dimensional feature vector, followed by dropout for regularisation and a fully connected layer that outputs a single logit corresponding to the probability of waste presence. The model is trained with the AdamW optimiser, and mixed-precision training is used.

To stabilise optimisation on this small and imbalanced dataset, we adopt a freeze–unfreeze schedule: during the first few epochs, only the FPN and classification head are trained while the backbone remains frozen; after this warm-up phase, all layers are unfrozen and fine-tuned jointly. Training loss, validation accuracy, F1 score, and ROC–AUC are evaluated at each epoch, and the checkpoint with the highest validation AUC is selected for downstream evaluation.

4.2. BBox Prediction

4.2.1. Data preparation and balancing

Bounding box prediction on our dataset presents several specific challenges. Objects of interest are usually small and sparsely distributed, and many images contain only background. In addition, the original data are organised into a limited number of "parent" scenes, from which multiple patches are extracted. If we were to split the data naively at patch level, patches from the same parent image could end up in different splits, effectively leaking scene information from train into validation and test.

The first step therefore focuses on dataset preparation and balancing. We enforce the split at *parent-image level* and then regenerate the patches so that each parent scene is assigned to exactly one split (train, validation or test). In this way, any patch derived from a given parent image can only appear in one split, so the model is never evaluated on regions of a scene it has already seen during training. This yields a stricter and more realistic generalisation test. During this process we compute simple statistics for each crop (number of objects and total annotated area). Patches with no boxes are treated as background-only; instead of keeping all of them, we cap their proportion so that background images remain present (to control false positives) but do not dominate the training set. Conversely, for rare classes we duplicate their patches within the training split and rely on copy-paste/mosaic augmentation to create additional synthetic examples, leading to a more balanced and informative training distribution.

4.2.2. Model choice

Given these constraints, we opted for a one-stage detector based on YOLOv8. Exploratory inspection and basic geometry statistics on the patch-based subset indicate that annotated waste instances are typically compact, with near-square aspect ratios, and appear as localised structures embedded in larger scenes. At the same time, the surrounding context is often visually cluttered: rubble piles, containers and bulky items can resemble rocks, construction materials or roofs when seen from above. YOLOv8's CSP-style backbone and PAN/FPN-style neck are designed to aggregate features across multiple spatial scales, which is crucial for retaining fine-grained detail on such localised objects while still capturing enough context to disambiguate clutter in overhead imagery (e.g. distinguishing waste piles from visually similar soil, roads or buildings). In addition, the anchor-free decoupled head produces dense predictions on high-resolution feature maps, making it well suited to our 896×896 patch-based tiling strategy and the small-to-medium object regime observed in the AerialWaste dataset.

From a practical standpoint, YOLOv8 offers an excellent trade-off between accuracy and computational efficiency for this setting. High-quality COCO/ImageNet-pretrained weights provide generic low-level texture and edge features that transfer well to aerial RGB imagery, even though the downstream classes are specific to waste. Compared to two-stage architectures (e.g. Faster R-CNN) or heavier transformer-based detectors, modern YOLO variants have been shown to deliver competitive performance on small-object detection in remote sensing while keeping inference cost manageable, which is important when large geographic areas must eventually be scanned via sliding-window inference. For these reasons we concentrated our effort on carefully tuning a single YOLOv8 family (backbone size, input resolution, loss weights and augmentation) rather than exploring many disparate architectures, focusing in particular on a medium-sized backbone at 896×896 resolution with strong but realistic data augmentation.

4.3. Segmentation

4.3.1. DeepLabV3 - ResNet-50

Semantic segmentation of satellite imagery presents numerous challenges. For instance, many objects are represented by very few pixels, and importantly, this means

that object boundaries are also represented in such a way. As such, segmentation of such imagery is sensitive to down-sampling methods. Moreover, images may exhibit low inter-class variation making it hard to distinguish between classes. As a general example, it may difficult to distinguish between the roof of a building and a parking lot. Therefore, we decided to use DeepLabV3 as our primary model.

By using atrous convolutions, DeepLabV3 [1] mitigates the impact of down-sampling of traditional CNN striding and pooling methods, ensuring the feature map preserves its resolution, allowing it to better identify boundaries. Furthermore, to address the second challenge mentioned previously, we require a model that can infer long range context. Another architectural feature of DeepLabV3 is that it uses Atrous Spatial Pyramid Pooling (ASPP) with global pooling which mitigates the impact of this issue by incorporating context from different distances away from an individual pixels local vicinity.

Additionally, we use a ResNet-50 backbone for this model due to its balance of size, computation and performance. By using skip connections, ResNet-50 can still learn features as the network gets deeper. Moreover, it is trained on ImageNet and hence, would be generalizable to many contexts, including the one of this paper.

4.3.2. Segment Anything Model (SAM)

Moreover, as previously mentioned, our dataset suffers from limited training data, hence, we also experiment with few-shot segmentation methods. More specifically, we use the Segment Anything Model (SAM) [2]. SAM utilizes a high-capacity Vision Transformer, capturing global context, mitigating the risk caused by inter-class similarity. The model is trained on over 1 billion masks, making it robust for few-shot classification.

5. Experiments

5.1. Binary Classification

The binary classification experiments constitute the first empirical component of our study. We begin with a baseline ResNet-50 classifier that uses only its deepest feature map and a global average pooling head. This baseline establishes a lower bound on expected performance and highlights the limitations of relying solely on single-scale features: specifically, the model struggles with small or low-contrast waste deposits, which often occupy only a limited portion of the scene.

Introducing the FPN head substantially improves performance without modifying the backbone or optimisation settings. On the held-out validation split, the final ResNet50+FPN classifier achieves a ROC–AUC of approximately 0.88 and a weighted F1 score of 0.76 at a threshold of 0.5. Class-specific metrics reveal the model's behaviour under strong class imbalance: the *Negative* class attains a precision of 0.98, a recall of 0.64, and an F1 score of 0.77, while the *Positive* class achieves a recall of 0.98 and an F1 score of 0.72 but with a lower precision of 0.57. This pattern reflects a deliberate modelling choice: prioritising high recall for waste-containing images to reduce the likelihood of missing hazardous sites, even at the cost of reduced precision. The overall accuracy of 0.75 is consistent with this conservative operating regime. Inspection of the predicted probability distributions further confirms clear class separation, with background images clustering near zero and positive images concentrated near one (see Figure 1).

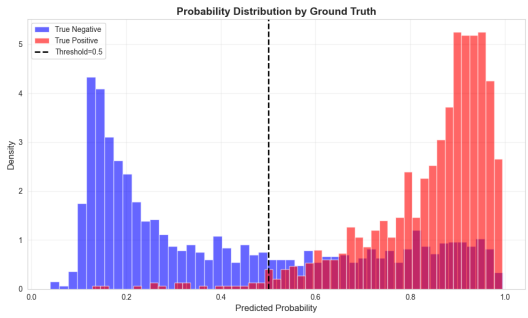


Figure 1. Probability distribution for Ground Truth Positives and Negatives

We additionally conducted ablation studies to assess the contribution of our modelling decisions. Removing either the positive-class weighting or the weighted sampler results in a substantial decline in recall for the Positive class, underscoring the necessity of explicit imbalance mitigation. Training with an unfrozen backbone from the outset similarly leads to noisier optimisation and slightly lower AUC, whereas the adopted freeze–unfreeze schedule yields more stable convergence and more consistent validation performance.

Overall, these experiments demonstrate that a moderately sized ResNet-50 backbone, equipped with an FPN and trained with targeted imbalance handling, provides an effective and computationally efficient foundation for image-level waste detection. The classifier achieves high sensitivity to waste presence while maintaining reasonable precision, and it serves as a strong high-level indicator that complements the more fine-grained localisation models introduced in the following sections.

5.2. BBox Prediction

The modelling notebooks follow an iterative design. We began with lightweight YOLOv8-small models at different input resolutions (640, 896 and 1024 px). These runs established baseline performance and highlighted the main failure modes: limited recall on small and visually subtle instances at low resolution, and increased false positives when simply upscaling the input without adjusting the architecture or training scheme. In a second step we switched to the medium backbone (YOLOv8m), which offers higher capacity while remaining feasible on a single GPU. At 896 px this configuration provided a clear improvement in detection quality, with higher recall and mAP compared to all small-backbone baselines.

The final version focuses on targeted refinements of this best configuration. We extended the training schedule (100 epochs with early stopping), adjusted the loss weights (in particular increasing the classification term relative to the defaults) to better reflect the binary waste-versus-background target, and tuned data augmentation with a combination of mosaic, mixup, copy-paste, colour jitter and light geometric transforms. These augmentations are particularly important in our setting because many waste patterns are rare and appear in only a handful of scenes; synthetic composition helps expose the model to more diverse contexts while keeping the test distribution unchanged.

Our final model, denoted `y8m_896_tuned3`, achieves an mAP50 of approximately 0.35 and an mAP50–95 of 0.18 on the evaluation split, with mean IoU of true positives around 0.75. This indicates that, when the detector fires, bounding boxes are typically well aligned with the ground truth, and that the main remaining challenges lie in recovering the hardest and rarest instances and in dealing with visually ambiguous background structures, rather than in pure localisation accuracy. Overall, the sequence of notebooks documents a coherent progression: from careful data splitting and dataset balancing, through baseline YOLOv8 experiments, to a tuned medium-backbone model that is well matched to the small, imbalanced and scene-structured nature of our dataset.

5.3. Segmentation

5.3.1. DeepLabV3 - ResNet-50

Following a series of experiments, we found that the hyper-parameters that delivered the best performance include a learning rate of 1×10^{-4}, a mini-batch of size 8, and a weight-decay of 1×10^{-5}. Moreover, we find that for the atrous convolutions of DeepLabV3, $6, 12, 18$ delivered the best performance, unchanged from the default configuration. However, it is worth noting that these figures are highly dependent on the subsample used to finetune this. In many experiments $3, 6, 12$ was chosen which we believe to be more plausible for this domain. This is due to the nature of satellite imagery where objects appear smaller than regular imagery. As a result, smaller contextual range is needed resulting in smaller dilations.

Furthermore, our training including a 10 epoch warm up period where only the DeepLabV3 head is being trained, with cross-entropy loss as the loss function. That is then followed up by progressive unfreezing of the ResNet50 backbone, for a total of 50 epochs of training. At each unfreeze, the learning rate is reduced by a factor of 10, to prevent overfitting.

Throughout the 50-epoch training schedule, the model optimised steadily.. The training loss decreased sharply from an initial value of approximately 0.48 to below 0.01 by the final epoch, while the validation loss remained within a consistent band despite the expected fluctuations typical of high-capacity segmentation networks. Validation IoU improved from an initial value of 0.3564 to a best value of 0.70 at epoch 47, and validation pixel accuracy steadily increased, eventually approaching 98%. These trends indicate stable learning without overfitting, even as progressively deeper layers of the backbone were unfrozen during fine-tuning.

However, as reflected in Table 1, the mean IoU is somewhat biased by the model's consistently strong performance on class 0, the background, whose IoU reached 0.9877 on the test set. Several foreground classes also achieved robust IoU scores, most notably dense classes 3, 4, 7, and 8, whereas others remained more challenging, such as dense classes 5 and 9, which exhibited lower IoUs (0.44 and 0.49 respectively). This behaviour is consistent with class-imbalance observed in the dataset, where the background dominates the pixel space. Moreover, because the ground truth masks are polygon-based rather than pixel-accurate, the mean IoU alone does not fully capture the visual correctness of predictions, and further qualitative inspection is warranted.

The plots in Figure 3 contextualise these results by illustrating how accuracy, loss, and IoU evolve jointly. Validation IoU and accuracy continue to improve even during intervals where validation loss temporarily increases, suggesting that the model is learning increasingly coherent structural features. The class-wise IoUs extracted from the logs highlight both the model's strengths on visually distinctive or frequent classes, and its difficulty on more heterogeneous or rare categories.

Finally, Figure 2 shows representative predictions on the unseen test set. The model achieves a test loss of 0.12, a mean IoU of 0.72, and a test accuracy of 0.98, confirming strong generalisation performance and consistent behaviour across train, validation, and test splits.



Figure 2. Example of DeepLabV3 Segmentation

5.3.2. Segment Anything Model (SAM)

For the SAM model, we tested several levels of few-shot learning, starting with zero-shot and ultimately settling with 100-shot due to superiour performance in mIoU, although 64 examples performed similarly well.

We implement a full few-shot fine-tuning and evaluation pipeline for the Segment Anything Model (SAM). We first load a small labeled dataset of landfill patches and freeze SAM's image encoder while fine-tuning only the mask-decoder using binary cross-entropy with logits on ground-truth masks. Each training sample is used in both its original and a lightly augmented form (brightness/contrast jitter) to improve generalization. During fine-tuning, SAM updates the decoder so that its predicted masks better align with the true segmentation. After training, we predict the remaining test images, generating SAM masks via bounding-box prompts and selecting the best-scoring prediction. Performance is then evaluated using a set of metrics including IoU, precision, recall, and F1.

Across the full test set of 2,900 images, the fine-tuned SAM model demonstrated strong localization performance, achieving an average IoU of 0.74 with a median of 0.77, indicating consistent segmentation quality across most samples. Precision and recall were also high, 0.79 and 0.92, respectively, reflecting SAM's tendency to slightly over-segment while still maintaining reliable boundary accuracy.

Although IoU values varied across the dataset, ranging from 0.08 to 0.94, the overall standard deviation of 0.13 suggests moderate variability likely tied to challenging edge cases or complex backgrounds. Because this task focuses solely on binary object segmentation given a bounding box, rather than multi-class prediction, the IoU distribution provides the primary lens for evaluation. Overall, the results show that the adapted SAM model performs robustly in this few-shot setting, outperforming the earlier DeepLabV3 baseline and demonstrating strong generalization despite limited training data.

Figure 5 shows an example of SAM's segmentation.

6. Conclusions

In this work, we have shown that modern computer vision pipelines can effectively support the discovery of illegal landfills from overhead imagery, even in the presence of scarce annotations and strong class imbalance. By combining image-level screening, object detection and segmentation within the EcoEyeSpy framework, we exploit complementary models to move from coarse scene understanding to precise localisation of suspicious waste sites. The use of foundation models such as Segment Anything further illustrates how generic segmentation priors can be adapted to a specialised environmental-monitoring context with minimal task-specific supervision. Overall, our results highlight the potential of deep learning and large-scale aerial datasets like AerialWaste to transform environmental monitoring into a more proactive, data-driven process. Future extensions could integrate temporal information, multimodal data and active learning to build even more robust and actionable tools for public authorities and policy makers.

EcoEyeSpy

Supplementary Material

7. Appendix

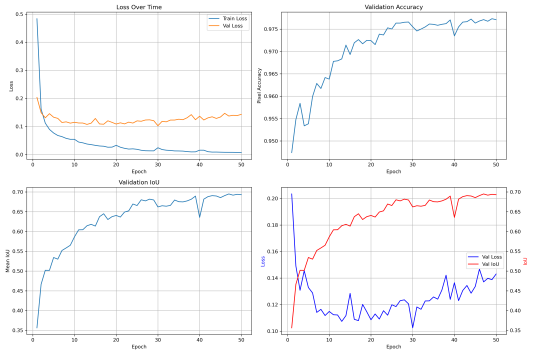


Figure 3. Training losses, accuracies, and IoU progression over 50 epochs.

Table 1. Per-class IoU values.

Class	IoU
0	0.9877
1	0.7555
2	0.6433
3	0.7205
4	0.8062
5	0.4414
6	0.6157
7	0.8254
8	0.8799
9	0.4912

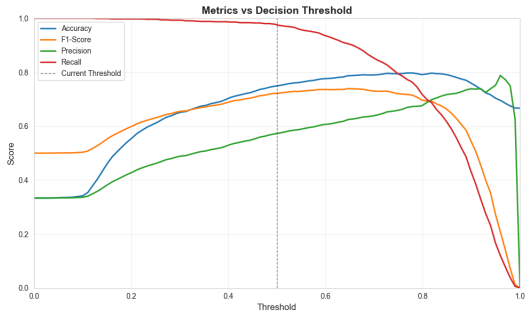


Figure 4. Performance for different thresholds



Figure 5. Example of SAM Segmentation

References

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